

Soru $|x\rangle = \frac{1}{\sqrt{2}} \left| \frac{1}{2}, \frac{1}{2} \right\rangle + \frac{1}{\sqrt{2}} \left| \frac{1}{2}, -\frac{1}{2} \right\rangle$ S_x operatörünün $\frac{\hbar}{2}$ i özdeğerli bir örnektir.

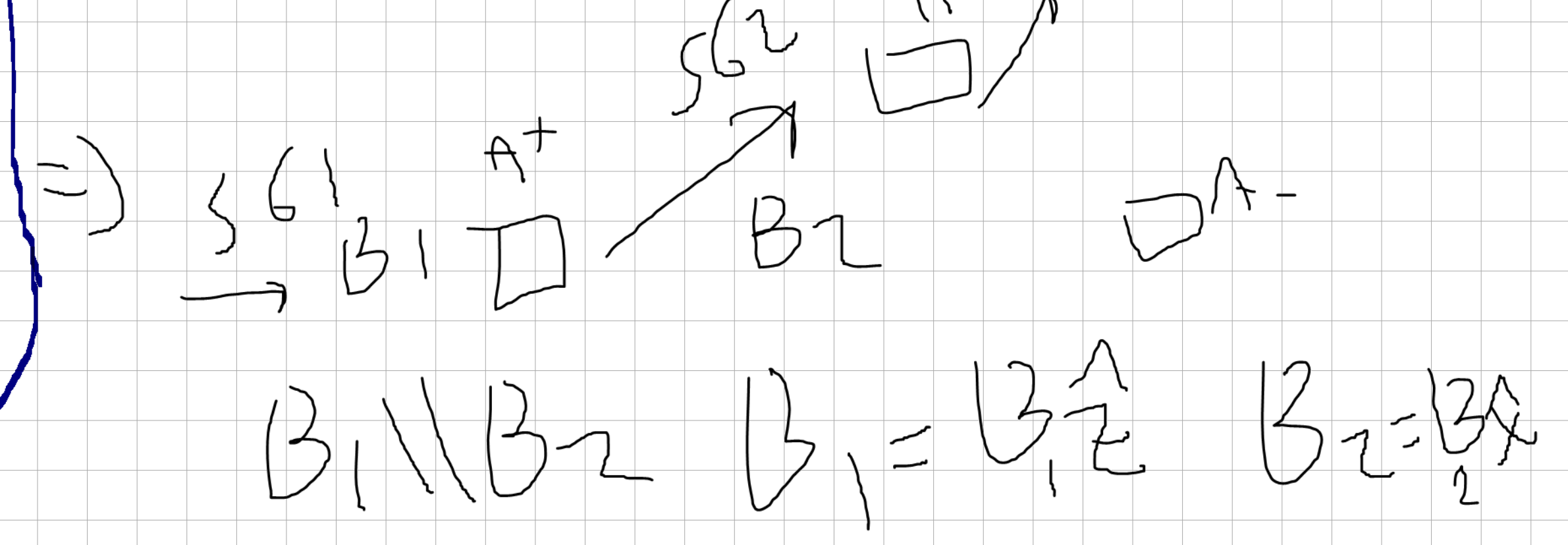
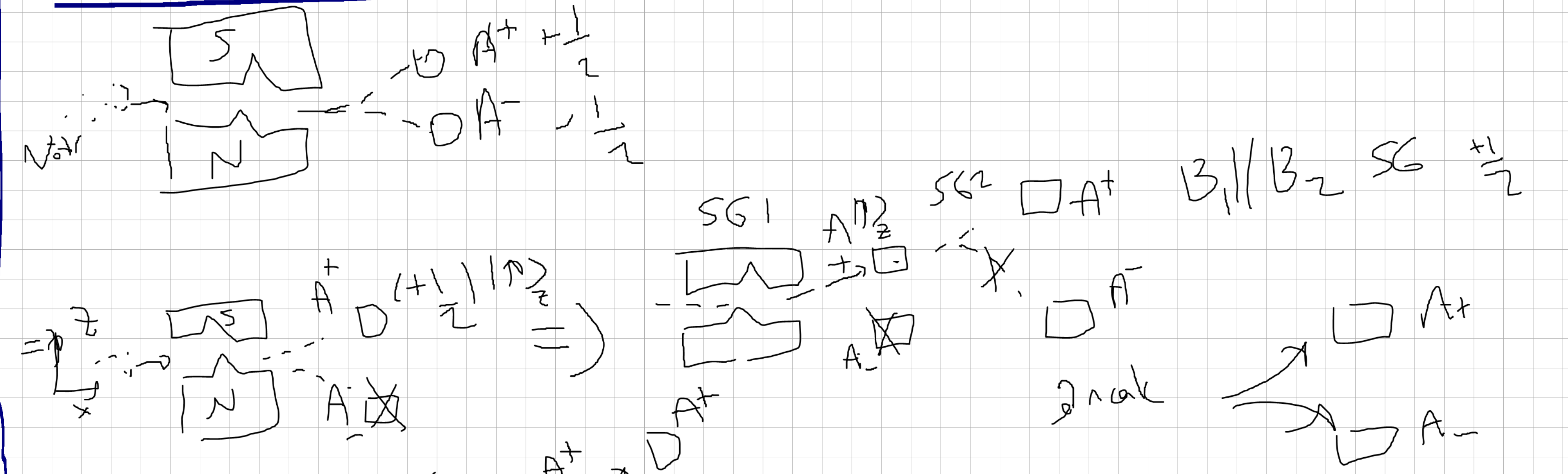
$|x\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle_z + |\downarrow\rangle_z)$
 $|x\rangle = \frac{1}{\sqrt{2}} (|0\rangle_z + |1\rangle_z)$

$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$S_x = \frac{\hbar}{2} \sigma_x \quad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

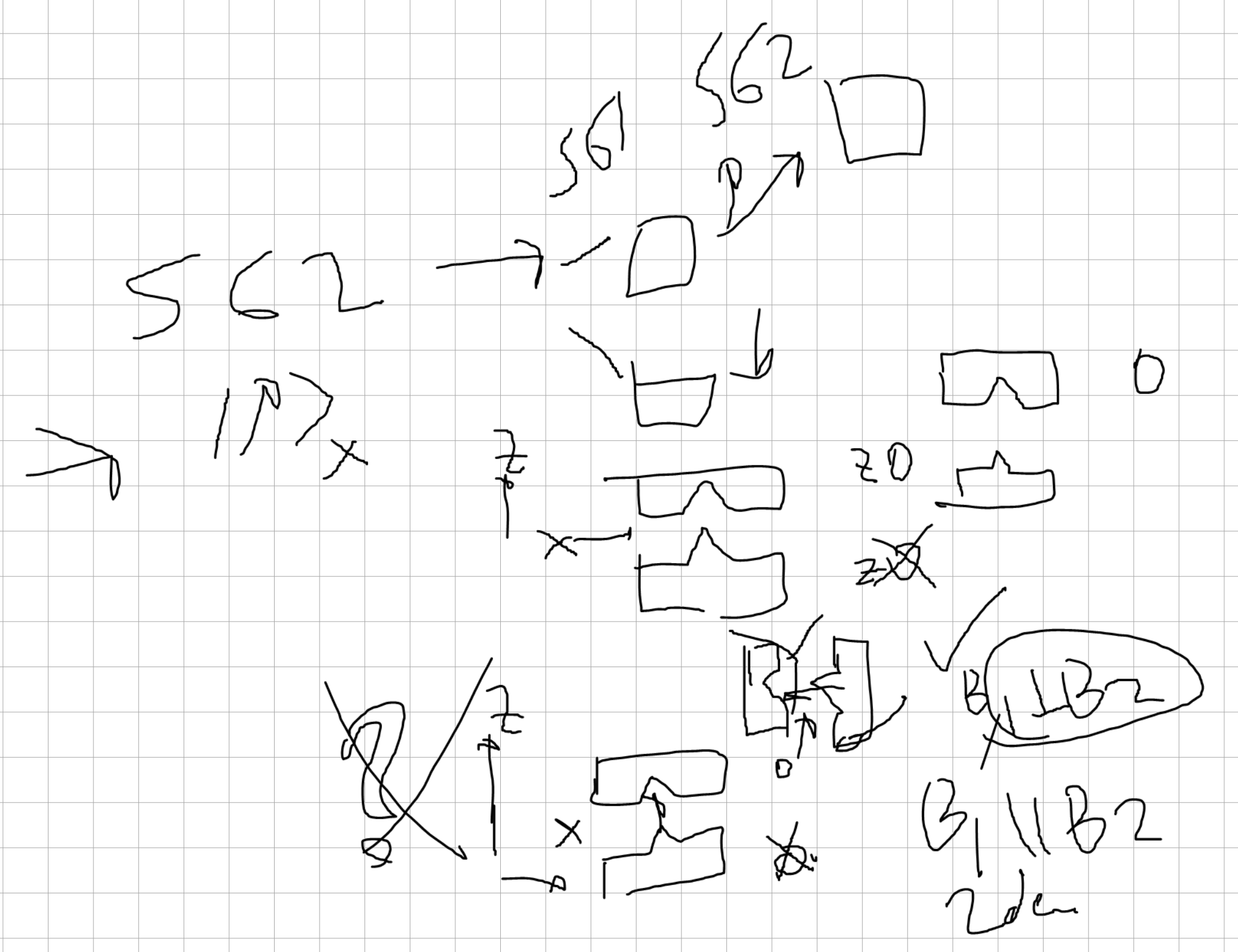
$S_x |x\rangle = \frac{\hbar}{2\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \frac{\hbar}{2\sqrt{2}} = \frac{\hbar}{2} |x\rangle$

Ardisik Stern-Gerlach Deneyi



$|x\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle_z + |\downarrow\rangle_z)$
 $|y\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle_z - |\downarrow\rangle_z)$

$|\uparrow\rangle_x + |\downarrow\rangle_x = \frac{2}{\sqrt{2}} |\uparrow\rangle_z \Rightarrow |\uparrow\rangle_z = \frac{1}{\sqrt{2}} (|\uparrow\rangle_x + |\downarrow\rangle_x)$
 $\begin{pmatrix} \sqrt{2} \\ 1 \end{pmatrix}$



$S_z |\uparrow\rangle_z = \frac{\hbar}{2} |\uparrow\rangle_z$
 $S_z |\downarrow\rangle_z = -\frac{\hbar}{2} |\downarrow\rangle_z$
 $\hat{L}_+ = L_x + iL_y$
 $\hat{L}_- = L_x - iL_y$
 $\hat{S}_+ = S_x + iS_y$
 $\hat{S}_- = S_x - iS_y$
 $S_+ + S_- = 2S_x \Rightarrow S_x = \frac{S_+ + S_-}{2}$
 $S_x = \frac{1}{2} (S_+ + S_-)$
 $\frac{1}{\sqrt{2}} (|\uparrow\rangle_z + |\downarrow\rangle_z)$
 $\Rightarrow \frac{1}{2\sqrt{2}} [S_+ |\uparrow\rangle_z + S_- |\uparrow\rangle_z + S_+ |\downarrow\rangle_z + S_- |\downarrow\rangle_z]$
 $\Rightarrow \frac{1}{2\sqrt{2}} \left[\hbar \sqrt{\frac{1}{2}(\frac{1}{2}+1)} \frac{1}{\sqrt{2}} |\downarrow\rangle_z + \hbar \sqrt{\frac{1}{2}(\frac{1}{2}-1)} \frac{1}{\sqrt{2}} |\uparrow\rangle_z \right]$
 $\Rightarrow \frac{\hbar}{2\sqrt{2}} [|\downarrow\rangle_z + |\uparrow\rangle_z] = \frac{\hbar}{2} |x\rangle$